ALICE L WALTON SCHOOL OF MEDICINE

# Antiphospholipid Antibody Syndrome

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Adapted from: Antiphospholipid Antibody Syndrome

Listen to this Brick
0:00 / 19:14 1x

Learning Objectives (4) Versions

***After completing this brick, you will be able to:***

1. Differentiate the clinical presentation of antiphospholipid antibody syndrome (APS).
2. Explain the mechanisms underlying APS.
3. Interpret cli[illegible]
4. Outline the management plan of APS.

CASE CONNECTION

MG, a 34-year-old female with systemic lupus erythematosus (SLE), is seeing you after a recent miscarriage. "We're heartbroken. This was my third miscarriage. I can't handle going through this again. Why is this happening?" You review past lab reports and find she has an activated partial thromboplastin time (aPTT) above the normal range. You think that this should cause bleeding complications, but then you remember that MG has SLE.

How will you explain what might be causing the miscarriages? How should her condition be managed? Consider your answers as you read, and we'll revisit MG at the end of the brick.

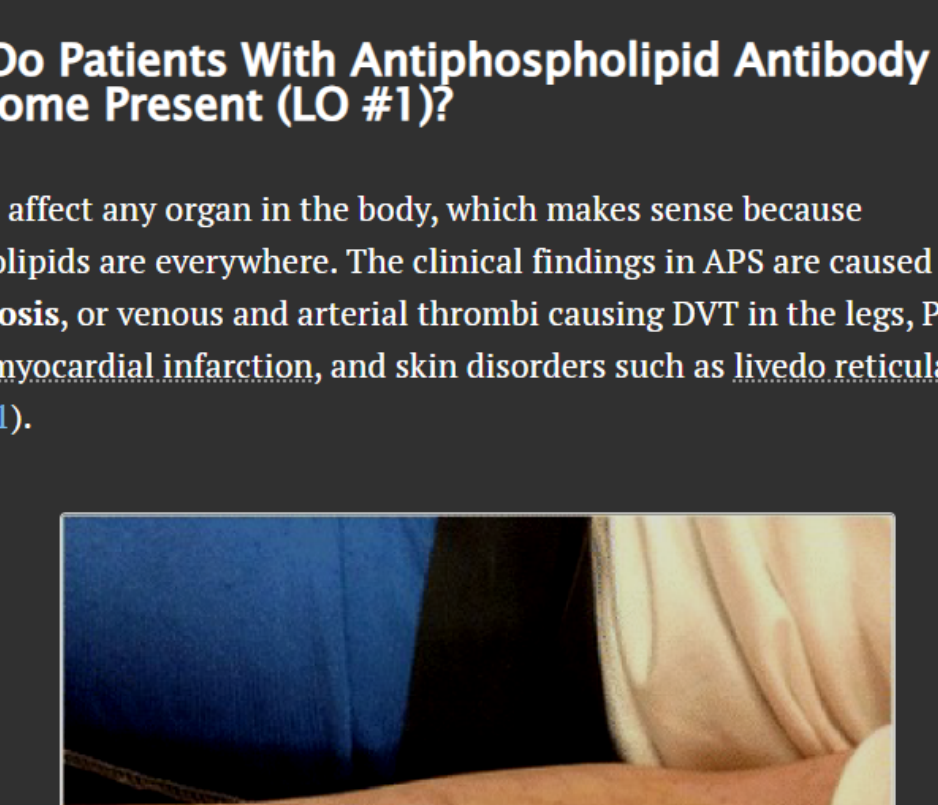
GO TO CONCLUSION ↓

## What Is Antiphospholipid Antibody Syndrome?

Antiphospholipid antibody syndrome (APS) is an autoimmune disorder characterized by the presence of prothrombotic circulating autoantibodies along with thromboembolic events such as deep venous thrombosis (DVT) and pulmonary emboli (PE). It is the most common acquired disorder of thrombosis. The annual incidence is estimated to be 2.1 per 100,000 population, and the estimated prevalence is 50 per 100,000 population. It is seen in an estimated 10% of patients with pregnancy losses. APS is associated with other autoimmu[illegible] **erythematosus** (**SLE**), 50% of the time.

## How Do Patients With Antiphospholipid Antibody Syndrome Present (LO #1)?

APS can affect any organ in the body, which makes sense because phospholipids are everywhere. The clinical findings in APS are caused by **thrombosis**, or venous and arterial thrombi causing DVT in the legs, PE, stroke, myocardial infarction, and skin disorders such as livedo reticularis (Figure 1).



Another characteristic feature of APS is **frequent fetal loss during pregnancy** from premature birth, placental insufficiency, and fetal death.

Any of these symptoms in a young person should raise concern. For example, some patients may present in their 20s with myocardial infarction or stroke in the absence of any family history of heart disease.

How does APS present?

## What Is the Pathophysiology of Antiphospholipid Antibody Syndrome (LO #2)?

APS is caused in part [illegible] alone does not usually cause disease.

### Antiphospholipid Antibodies

Recall that the phospholipid bilayer is the universal component of cell membranes. Antiphospholipid antibodies, as the name implies, are a group of antibodies that attack phospholipids, including those in cell membranes. Although this sounds dangerous, these antibodies do not always cause significant harm to the patient.

The exact mechanism of how antiphospholipid antibodies may cause excess thrombosis is unclear, but the antibodies seem to alter the coagulation pathway in various ways that promote hypercoagulability. Again, not everyone with antiphospholipid antibodies develops blood clots—up to 5% of healthy individuals are estimated to have circulating antiphospholipid antibodies. But, once someone with antiphospholipid antibodies starts to have symptoms such as recurrent thrombosis or fetal loss, APS should be suspected.

### Pathophysiology

Because the presence of antiphospholipid antibodies alone does not lead to APS, researchers have proposed a "two-hit" theory in which at least two components are required for the full-blown manifestation of a disease. In APS, the first component is the presence of antiphospholipid antibodies. On top of this, any additional risk factor for hypercoagulability, such as smoking, surgery, diabetes, pregnancy, oral contraceptive intake, and malignancy, could deliver the second hit, resulting in the development of APS.

How does the two-hit theory apply to APS?

### Causes

In 50% of patients, the formation of antiphospholipid antibodies is idiopathic.

CLINICAL CORRELATION

However, some systemic diseases—most commonly, **systemic lupus erythematous** (**SLE**)—predispose patients to their development. In fact, approximately 20% to 40% of patients with SLE have antiphospholipid antibodies, and this is why one of these is named **lupus anticoagulant**. The confusing name (because the patient's problem is actually too much coagulation) originates from the laboratory observation that it delays coagulation in vitro.

Additional causes of [illegible] drugs, infections, or some idiopathic autoimmune disease with no underlying cause detected. Of note, antiphospholipid antibodies due to infections are usually related to a viral infection and are transient (no clinical consequence).

## How Do We Diagnose Antiphospholipid Antibody Syndrome (LO #3)?

Diagnosis of APS is made by both clinical and laboratory analyses. In general, test patients who have thrombi and are young or lack other risk factors (e.g., pregnancy, oral contraceptives, surgery, cancer, long air travel). Also, consider testing patients with one or more unexplained episodes of pregnancy loss beyond 10 weeks or three or more spontaneous abortions before 10 weeks.

Once you decide to test, labs indicating APS include the presence in serum of one or more of:

- Lupus anticoagulant
- Anticardiolipin antibodies
- Anti-β-2-glycoprotein-I antibodies

CLINICAL CORRELATION

The lab abnormalities must be confirmed on **2 occasions, at least 12 weeks apart.** This is to exclude the transient, insignificant antiphospholipid antibodies that represent the majority of cases. Someone could have a DVT or stroke and coincidentally have a transient, totally unrelated antiphospholipid antibody; this would not represent [illegible]

Other blood abnormalities in these patients may include **low platelet count** (resulting from platelet consumption in the clots), **low serum complement** (resulting from complement consumption in immune complex formation as part of the autoimmune response), and a **prolonged activated partial thromboplastin time** (**aPTT**).

The prolonged aPTT may sound puzzling at first, because it seems paradoxical that aPTT would be prolonged in a person who is in a hypercoagulable state. We see this unique phenomenon in APS because antiphospholipid antibodies bind to and inhibit the reagent (thromboplastin) used in the coagulation studies. This means that the antibodies act to inhibit coagulation in vitro, whereas they may act as procoagulants in vivo.

Why is the aPTT sometimes prolonged in APS?

## How Do We Treat Antiphospholipid Antibody Syndrome (LO #4)?

Patients with APS and thrombi (DVT, PE) are treated with anticoagulants, just like any patient with a thrombus. After treatment, patients with thrombi should be kept on long-term anticoagulation because of the risk of re-thrombosis. Most physicians use warfarin for prophylaxis.

For females with pregnancy loss who are diagnosed with APS, long-term prophylactic anticoagulation may be considered in certain cases. If these patients become pregnant again, anticoagulation with low-molecular-weight heparin and low-dose aspirin are often recommended to try to avoid recurrent loss of the pregnancy.

How do we prevent rethrombosis in patients with prior thrombosis?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to MG, how do you explain what might be causing her miscarriages? How do you proceed with her treatment?

After further testing, you tell MG that she has antiphospholipid antibody syndrome (APS), which is seen in patients with SLE. APS causes blood clots to form, increasing the chance of recurrent miscarriages. You tell MG that you want to refer her to a high-risk pregnancy specialist if she decides to become pregnant again, because some treatment options (such as anticoagulation) decrease the risk of miscarriages. "If there is treatment that will give us a better chance to have a child, I'd like to hear more about it," she says.

## Summary

- Antiphospholipid antibody syndrome (APS) is an autoimmune disorder characterized by the presence of antiphospholipid antibodies along with thromboembolic events and/or failed pregnancies.
- Antiphospholipid antibodies are a group of antibodies that target phospholipids and phospholipid-related proteins.
- Clinical findings are caused by thrombosis and include deep vein thrombosis, pulmonary emboli, stroke, myocardial infarction, skin disorders, and miscarriage.
- Patients may have deep vein or arterial thrombi without obvious risk factors.
- Formation of antiphospholipid antibodies is associated with autoimmune diseases such as [illegible] In 50% of patients, the formation of [illegible]pathic, but systemic diseases such as SLE predispose patients to their development.
- The "two-hit" theory of APS includes the presence of antiphospholipid antibodies along with any additional risk factor for hypercoagulability (e.g., smoking, surgery, diabetes, pregnancy, oral contraceptive intake, and malignancy).
- Diagnosis is based on clinical presentation and laboratory findings.
- Patients with a history of pregnancy losses, unexplained thrombi, or thrombotic events in younger patients should be tested for APS.
- The presence of antiphospholipid antibodies such as lupus anticoagulant, anticardiolipin, or anti-β-2-glycoprotein-I antibodies, confirms APS. These tests must be done on 2 separate occasions, at least 12 weeks apart.
- Other laboratory findings in patients with APS can be low platelet count, low serum complement, and a prolonged activated partial thromboplastin time (aPTT).
- Patients with APS and thrombi should be treated with standard anticoagulation, then long-term warfarin therapy.
- Pregnant patients with APS are not usually treated with long-term anticoagulation, but treatment with low-molecular-weight heparin and low-dose aspirin during future pregnancies is recommended.

## Review Questions

**1. A 25-year-old female has recently been diagnosed with antiphospholipid antibody syndrome. Which of the following additional conditions is this woman most likely to have?**

A. ○ Factor V [illegible]
B. ○ Multiple myeloma
C. ○ Protein C deficiency
D. ● Systemic lupus erythematous

**Hide Explanation**

The correct answer is systemic lupus erythematous (D) (SLE). SLE is the most common cause of secondary antiphospholipid antibody syndrome. Up to 20% to 40% of patients with SLE demonstrate antiphospholipid antibodies. Factor V Leiden (A) is a hereditary thrombotic disorder. Multiple myeloma (B) is neoplastic proliferation of plasma cells, which results in excessive production of monoclonal antibodies. Protein C deficiency (C) is another hereditary thrombotic disorder.

**2. A 31-year-old female with a history of deep vein thrombosis and two miscarriages who is otherwise healthy presents to the clinic. Which of the following laboratory findings would be most consistent with a diagnosis of antiphospholipid antibody syndrome?**

A. ● Elevated activated partial thromboplastin time
B. ○ Elevated international normalized ratio
C. ○ Leukocytosis
D. ○ Normal partial thromboplastin time
E. ○ Thrombocytosis

**Hide Explanation**

The correct answer is elevated activated partial thromboplastin time (A). Antiphospholipid antibody syndrome (APS) is not associated with elevated international normalized ratio (B) or leukocytosis (C). APS is associated with prolonged partial thromboplastin time (PTT)/prothrombin time (PT), not normal partial thromboplastin time (D); and with thrombocytopenia, not thrombocytosis (E). Remember that antiphospholipid antibodies act to inhibit coagulation in vitro, whereas they act as procoagulants in vivo.

**3. Which of the following is the most likely finding in a patient with antiphospholipid antibody syndrome?**

A. ○ Epistaxis
B. ○ Gum bleeding
C. ● Livedo reticularis
D. ○ Palmar erythema
E. ○ Spider angiomas

**Hide Explanation**

The correct answer is livedo reticularis (C). A clinical finding of antiphospholipid antibody syndrome (APS) is livedo reticularis, a mottled pattern of skin discoloration. This condition is caused by obstruction of superficial capillaries by small blood clots. Epistaxis (A) and gum bleeding (B) are associated with bleeding disorders, such as conditions with clotting factor deficiency. Palmar erythema (D) and spider angiomas (E) have been associated with liver disease, not APS.

**4. Which of the following statements is most accurate about the "two-hit" theory for antiphospholipid antibody syndrome (APS)?**

A. ○ Antiphospholipid antibodies are the only component of APS pathophysiology.
B. ○ Factors such as smoking, surgery, diabetes, pregnancy, oral contraceptive intake, and malignancy can prevent the full manifestation of APS.
C. ● The presence of antiphospholipid antibodies alone does not lead to the full manifestation of APS.
D. ○ Two incidents of blunt force trauma can lead to the development of APS.
E. ○ Two or more risk factors for thrombosis are required for the diagnosis of APS.

**Hide Explanation**

The correct answer is the presence of antiphospholipid antibodies alone does not lead to the full manif[illegible]dies are not the only component of APS pathophysiology (A); the theory states that an additional factor must be present. Factors such as smoking, surgery, diabetes, pregnancy, oral contraceptive intake, and malignancy can prevent the full manifestation of APS (B) is incorrect. These factors denote additional risk factors for APS, which can act as the second "hit" in APS disease pathophysiology. Two incidents of blunt force trauma can lead to the development of APS (D) is unrelated and false. Two or more risk factors for thrombosis are required for the diagnosis of APS (E) is incorrect. The theory does not specify how many additional conventional risk factors are necessary for the full development of APS.

**5. A patient with a recent diagnosis of antiphospholipid antibody syndrome presents to the hospital with a deep venous thrombosis (DVT). What is the proper treatment for their condition?**

A. ● Anticoagulation
B. ○ Desmopressin (DDAVP)
C. ○ Factor replacement
D. ○ Platelet transfusion

**Hide Explanation**

The correct answer is anticoagulation (A), with heparin and warfarin. Desmopressin (DDAVP) (B) is a treatment typically used for bleeding disorders, such as von Willebrand disease and hemophilia. Factor replacement (C) is a treatment typically used for bleeding dis[illegible]ent for disorders involving platelets.

**How would you rate this Brick?**

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**Review Learning Objectives**

Check off the objectives when you feel comfortable with the concept

01 ☐ Differentiate the clinical presentation of antiphospholipid antibody syndrome (APS).
02 ☐ Explain the mechanisms underlying APS.
03 ☐ Interpret clinical findings to properly diagnose APS.
04 ☐ Outline the management plan of APS.

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**Objective Confidence: 0/4**